Quantum-Assisted Robust Maneuver Initialization for Low-Thrust Cislunar Transfers Under Missed-Thrust Events

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Abstract

Low-thrust cislunar maneuver design couples nonlinear continuous trajectory optimization with discrete choices about when the spacecraft thrusts, coasts, and recovers after outages. This paper investigates a quantum-ready initialization architecture in which binary optimization proposes coarse thrust/coast schedules and a classical optimizer refines norm-bounded piecewise-constant thrust vectors, including selected missed-thrust recovery branches. The study is deliberately framed as initialization rather than replacement of deterministic optimal control. A robust binary objective is evaluated by Earth–Moon Circular Restricted Three-Body Problem propagation, includes nominal terminal error, outage terminal error, degradation, propellant use, switching penalties, and an optional duty-cycle prior, and is approximated by a quadratic unconstrained binary optimization (QUBO) surrogate for simulated annealing and statevector QAOA sampling. The paper reports ten evidence components: a hard catalog-derived NRHO-like to DRO-phase benchmark that remains infeasible under the current thresholds, feasibility and row-truthful direct multiple-shooting stress tests showing that this hard case is beyond the present refinement capability, a stronger targeted catalog feasibility attempt that also fails, bounded-projected multiple-shooting diagnostics on non-teacher catalog halo phase-shift cases, a compact bounded trapezoidal direct-collocation baseline on the same phase-shift suite, a non-teacher catalog halo phase-shift benchmark that is solved by an all-windows continuous baseline but not by the tested binary schedule samplers, a deterministic dense-schedule repair ablation for that phase-shift case, a non-teacher 12-segment cardinality-prior phase-shift benchmark, an optimized-angle QAOA depth ablation, and a disclosed teacher-controlled benchmark whose target is generated by a bounded low-thrust trajectory. In the bounded-projected phase suite, two of three multiple-shooting selected-branch rows satisfy the configured thresholds; the direct-collocation baseline succeeds on the phase-time 0.4 row with nominal error 0.0357, selected-recovery error 0.0191, and all-mask diagnostic worst error 0.0602, but fails at phase times 0.2 and 0.3. On the unrepaired phase-shift sampler benchmark, the all-windows continuous baseline succeeds in 5/5 seeds, while all tested discrete initializers fail under the same thresholds. A prefix-to-last-active repair makes every discrete initializer succeed in 5/5 seeds, but only by converting sparse candidate schedules to the same all-windows refined schedule, `11111111`. In the 12-segment cardinality-prior benchmark, cross-entropy, genetic search, true-objective simulated annealing, and surrogate-QUBO simulated annealing each succeed in 10/10 seeds at equal 120 true-evaluation budgets and find one-coast schedules with median nominal fuel 0.0917, compared with 0.1000 for the all-windows baseline; random-angle QAOA in the original run succeeds in 4/10 seeds and random search in 2/10. A follow-on ablation using optimized QAOA angles on the fitted QUBO improves QAOA success to 6/10 for $p = 1$ and 9/10 for $p = 2$, matching exact top-24 QUBO candidate evaluation but not surrogate-QUBO simulated annealing at 10/10. The all-windows baseline still has lower terminal errors. On the teacher-controlled benchmark, selected-branch recovery succeeds for random search in 2/3 seeds and for cross-entropy, QUBO simulated annealing, and QAOA in 1/3 seeds each. These results identify duty-cycle-aware binary objectives, continuous-control initialization, recovery parameterization, and QUBO-to-schedule generalization as the dominant bottlenecks that must be solved before a Q1 aerospace-journal performance claim would be defensible.

Keywords: Low-thrust trajectory optimization, quantum annealing, QAOA, QUBO, cislunar dynamics, missed-thrust events, CR3BP

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1. Introduction

Electric low-thrust propulsion enables propellant-efficient cislunar and interplanetary transfers, but it also turns maneuver planning into a long-horizon nonlinear optimal-control problem. A practical low-thrust design must choose continuous thrust magnitudes and directions while also choosing a qualitative maneuver structure: which time intervals should thrust, coast, or recover from an outage. The continuous part is usually handled by direct transcription, collocation, multiple shooting, sequential convex programming, or indirect methods. The discrete structure is rarely searched exhaustively, yet it strongly affects whether a deterministic optimizer converges.

This initialization sensitivity is especially important for robust low-thrust design. Missed-thrust events caused by safe modes, propulsion interruptions, or operational constraints can invalidate an otherwise efficient transfer. Grebow and McCarty show that including missed-thrust branches in low-thrust cislunar design can greatly reduce recovery propellant for week-scale outages [1]. Sinha and Beeson identify initial-guess generation as a bottleneck for low-thrust trajectory design with missed-thrust robustness [2]. These results motivate the central question of this paper:

> Can a quantum-ready binary optimization layer generate useful robust low-thrust maneuver schedules when missed-thrust fragility is included in the binary energy model and the resulting schedules are refined by classical continuous optimization?

The question is timely because low-thrust trajectory optimization has recently been mapped into quantum-ready binary formulations. De Grossi et al. transcribed an Earth–Mars low-thrust trajectory optimization problem into a binary/QUBO form and tested D-Wave quantum and hybrid solvers, while emphasizing present hardware limits [3]. Moretta and Casasola studied quantum-assisted initialization for Earth–Moon low-thrust trajectories using binary thrust-activation patterns as seeds for deterministic solvers [4]. The present paper differs by making missed-thrust recovery part of the initialization objective and by reporting negative and controlled-feasible evidence side by side.

The paper makes four contributions. First, it formulates a robust binary thrust-window initialization problem in the Earth–Moon CR3BP with explicit missed-thrust outage masks. Second, it fits a QUBO surrogate to trajectory-evaluated robust costs and samples it with classical QUBO simulated annealing and a statevector QAOA simulator. Third, it connects each binary schedule to a norm-bounded branch-recovery least-squares refinement in which selected outage branches receive post-outage recovery controls. Fourth, it reports a reproducible evidence hierarchy rather than a single favorable benchmark: a hard catalog-derived NRHO-like to DRO-phase case, feasibility/multiple-shooting stress tests, bounded non-teacher phase-shift diagnostics including trapezoidal direct collocation, deterministic dense-schedule repair, duty-cycle-prior and optimized-QAOA ablations, and a fully disclosed teacher-controlled feasible case.

The claims are intentionally narrow. The experiments do not show quantum advantage, do not solve high-fidelity mission design, and do not demonstrate universal missed-thrust robustness. They test whether a quantum-ready discrete layer can be integrated into a robust low-thrust initialization pipeline, whether the continuous backend can solve any non-teacher cislunar case, and where the present pipeline fails.

2. Related Work

2.1. Low-thrust trajectory optimization

Low-thrust trajectory optimization has a mature literature spanning indirect methods, direct transcription, collocation, global search, and hybrid approaches. Morante et al. survey low-thrust optimization as a hybrid optimal-control problem with both continuous and discrete structure [5]. Direct methods convert the problem into a large nonlinear program, but their convergence and final local optimum can be sensitive to the supplied trajectory and control initialization. Indirect methods can be highly efficient near a solution but introduce boundary-value sensitivity through costates.

For preliminary cislunar design, the Circular Restricted Three-Body Problem is widely used because it captures the nonlinear Earth–Moon dynamical environment while keeping algorithmic studies tractable. The NASA/JPL Three-Body Periodic Orbits API provides normalized periodic-orbit states, periods, Jacobi constants, stability indices, and system constants for families such as halo orbits and distant retrograde orbits [6]. The catalog-derived benchmark in this paper uses those states as transparent boundary-condition seeds.

2.2. Missed-thrust robustness

Missed-thrust design has been studied through margin analysis, expected-thrust formulations, virtual recovery trajectories, robust optimal control, and branching trajectory methods. Rubinsztejn et al. embed expected thrust fraction into deterministic trajectory design and report improved resilience to missed-thrust scenarios [7]. Venigalla et al. formulate multi-objective low-thrust trajectory optimization with missed-thrust recovery margin [8].

Grebow and McCarty specifically study low-thrust cislunar transfers, including NRHO-to-DRO transfers, and optimize branching trajectories for outage recovery [1]. Sinha and Beeson focus on initial-guess generation for robust low-thrust missed-thrust design and compare conditional and nonconditional global-search strategies [2]. Robust cislunar low-thrust optimization has also advanced through sequential convex programming and covariance steering; Kumagai and Oguri use chance-constrained covariance steering to design nominal trajectories and correction policies under uncertainty in the Earth–Moon CR3BP [9]. These works set a high bar for any initialization method: a useful discrete initializer must ultimately feed a continuous optimizer that can produce robust feasible trajectories.

2.3. Quantum-ready binary optimization

Quantum annealers and QAOA target discrete optimization models such as Ising Hamiltonians and QUBOs. A QUBO minimizes

$$E(x) = c + \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j, \qquad x_i \in \{0, 1\}, \tag{1}$$

which is convertible to the Ising form used in quantum annealing and many gate-model workflows [10, 11]. Farhi et al. introduced QAOA as a gate-model variational algorithm for approximate combinatorial optimization [12]. Shukla and Vedula formulated trajectory optimization as a discrete search problem and argued that quantum search can provide speedups for suitably discretized cases [13]. De Grossi et al. provide the closest low-thrust QUBO precedent [3]. This paper uses QUBO and QAOA only for the discrete initialization layer; the continuous trajectory remains a classical optimal-control problem.

2.4. Cislunar initial guesses and direct transcription

The targeted catalog benchmark in this paper is deliberately harder than the teacher-controlled case because cislunar transfers between stable or nearly stable periodic orbits are known to be sensitive to the continuous initial guess. Pritchett, Howell, and Grebow use collocation, trajectory stacking, orbit chaining, and continuation to compute low-thrust transfers in the restricted three-body problem, including NRHO-to-DRO examples [14]. Their results are important context for this paper: binary thrust-window sampling is not a substitute for the trajectory-stacking, natural-arc, continuation, and direct-transcription machinery that mature cislunar low-thrust solvers use to enter a feasible basin.

3. Problem Formulation

3.1. Forced CR3BP dynamics

The spacecraft state is

$$s = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top \tag{2}$$

in the normalized Earth–Moon rotating frame. With mass parameter μ, distances to the primary and secondary bodies are

$$r_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}, \tag{3}$$

$$r_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}. \tag{4}$$

The forced CR3BP equations are

$$\ddot{x} = 2\dot{y} + x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3} + a_x, \tag{5}$$

$$\ddot{y} = -2\dot{x} + y - \frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3} + a_y, \tag{6}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3} + a_z. \tag{7}$$

The transfer duration T is divided into N uniform segments. A binary variable $x_i \in \{0, 1\}$ gates thrust availability during segment i. During the discrete search stage, a deterministic target-seeking steering law maps the binary schedule to a propagated trajectory,

$$g(s) = K_r(r_f - r) + K_v(v_f - v), \qquad a_i(s) = x_i a_{\max} \frac{g(s)}{\|g(s)\| + \epsilon}. \tag{8}$$

This steering law is not claimed to be optimal. It creates a repeatable map from a bitstring to a trajectory so that discrete schedules can be ranked before continuous refinement. A separate oracle diagnostic, introduced only for the teacher-controlled benchmark, initializes refinement from the hidden continuous controls used to generate the target. That diagnostic is not a competing initializer; it tests whether the refinement formulation can recover the known feasible trajectory when continuous-control initialization is no longer the bottleneck.

3.2. Missed-thrust masks and robust binary cost

Let $\mathcal{M}$ denote a set of outage masks. Each mask zeros thrust in a one- or two-segment contiguous window. For a nominal schedule x, the outage-realized schedule is $x \odot m$. The scaled terminal error under mask m is

$$e_m(x) = \left\| \begin{bmatrix} (r_T(x \odot m) - r_f)/\ell_r \\ (v_T(x \odot m) - v_f)/\ell_v \end{bmatrix} \right\|_2. \tag{9}$$

The trajectory-evaluated robust binary objective is

$$\begin{aligned} J(x) = {} & w_n e_0(x) + w_w \max_{m \in \mathcal{M}} e_m(x) + w_d \left(\max_{m \in \mathcal{M}} e_m(x) - e_0(x) \right) \\ & + w_p \frac{1}{N} \sum_{i=1}^{N} x_i + w_s \frac{1}{N-1} \sum_{i=1}^{N-1} |x_{i+1} - x_i|. \end{aligned} \tag{10}$$

The terms penalize nominal miss distance, worst outage miss distance, missed-thrust degradation, active thrust fraction, and thrust/coast switching.

4. Quantum-Assisted Initialization Method

4.1. QUBO surrogate

Because $J(x)$ depends on nonlinear propagation and outage evaluation, it is not itself a QUBO. The method therefore fits

$$\widehat{J}(x) = c + \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j \tag{11}$$

from trajectory-evaluated samples $\{x^{(k)}, J(x^{(k)})\}_{k=1}^{K}$ using ridge-regularized least squares:

$$\min_\theta \sum_{k=1}^{K} \left(J(x^{(k)}) - \widehat{J}(x^{(k)}; \theta) \right)^2 + \lambda \|\theta\|_2^2. \tag{12}$$

The fitted QUBO is then sampled by classical QUBO simulated annealing or converted to a diagonal cost Hamiltonian for statevector QAOA. The present experiments use QAOA only as a small simulated gate-model sampler; no hardware quantum advantage is implied.

4.2. Discrete initializers

The experiments compare random search, cross-entropy search, a genetic algorithm, true-objective simulated annealing, surrogate QUBO simulated annealing, and statevector QAOA. Random search and true-objective simulated annealing evaluate propagated schedules directly. Cross-entropy updates Bernoulli segment probabilities from elite samples. The genetic algorithm uses elitism, crossover, and mutation. The QUBO and QAOA methods use true trajectory evaluations to train $\widehat{J}$, then validate their best sampled schedules with the true objective.

Budget accounting is therefore reported in two ways: solver true evaluations exclude shared QUBO training, while total true evaluations include QUBO training. Equal total-evaluation comparisons are the relevant fair comparison.

4.3. Continuous branch-recovery refinement

The refinement stage converts a selected binary schedule into a continuous low-thrust candidate by optimizing piecewise-constant acceleration vectors only in active windows. Refined accelerations are projected to the Euclidean thrust ball $\|u_i\|_2 \leq a_{\max}$ before propagation and persistence. The branch-recovery mode optimizes a nominal control sequence plus selected-outage-specific post-outage recovery controls. Its least-squares residual includes nominal terminal error, selected outage terminal error, fuel regularization, control regularization, and inter-segment smoothness. A run is counted as successful only if both the nominal terminal error and the selected missed-thrust worst error are below the configured thresholds.

The paper also reports an all-mask diagnostic: after refinement on selected outages, every one- and two-segment outage mask is evaluated without claiming that all masks were optimized. This separates selected-branch recovery success from universal outage robustness.

5. Experimental Setup

5.1. Common dynamics and source state

All experiments use normalized Earth–Moon CR3BP dynamics with $\mu = 0.01215058560962404$. The common initial state is taken from the first row of an Earth–Moon L2 southern halo query with period in $[1.55, 1.65]$ TU from the NASA/JPL periodic-orbit API:

$$s_0 = [1.0324985738, 0, -0.1884844917, 0, -0.1249781287, 0]. \qquad (13)$$

This state is NRHO-like in the sense that it lies in the L2 southern halo family, but the paper treats it as a normalized benchmark state, not a certified operational NRHO.

5.2. Hard catalog-derived benchmark

The hard benchmark uses a distant retrograde orbit seed from an Earth–Moon DRO query with period in $[2.0, 3.0]$ TU and Jacobi constant in $[2.9, 3.1]$:

$$s_{\mathrm{DRO}} = [0.8161260827, 0, 0, 0, 0.5076556931, 0]. \tag{14}$$

The fixed target is generated by ballistic propagation of this DRO seed for the transfer time. The main catalog-derived run uses $T = 2.7$ TU, $N = 12$, $a_{\max} = 0.055$, one- and two-segment outage masks, two pilot seeds, and threshold values selected before the feasibility sweep. This case is intentionally retained even though it is hard for the current pipeline, because it prevents the paper from reporting only a constructed favorable target.

5.3. Catalog halo phase-shift benchmark

To separate continuous-backend feasibility from teacher-generated feasibility, the repository also includes non-teacher phase-shift benchmarks derived from the same JPL L2 southern halo source state. The target is generated by zero-thrust CR3BP propagation of s_0 for a phase time of 0.3 TU, while the transfer is required to arrive at that target after $T = 0.5$ TU. Thus the target is catalog/dynamics-derived rather than produced by a low-thrust teacher, but the spacecraft must correct a phase mismatch. The first configuration uses $N = 8$, $a_{\max} = 0.2$, one-segment outage masks, and the same nominal and selected robust thresholds as the other experiments. A zero-thrust trajectory has normalized nominal terminal error 0.3758, so the case is not a do-nothing pass. This benchmark is easier than the NRHO-like to DRO-phase transfer and is included to test whether the present continuous refinement can solve at least one non-teacher cislunar case.

The second phase-shift configuration uses $N = 12$ and adds a duty-cycle prior to the binary objective:

$$J_\rho(x) = w_\rho \left(\frac{1}{N} \sum_{i=1}^{N} x_i - \rho \right)^2, \tag{15}$$

with $\rho = 11/12$ and $w_\rho = 12$. This prior is not an oracle trajectory correction. It encodes a mission-design preference for one coast window while retaining a dense low-thrust arc, and it is included because the unrepaired $N = 8$ objective over-rewards sparse late thrust before continuous refinement.

5.4. Feasibility and multiple-shooting stress tests

To test whether the hard catalog-derived failure is mainly a search issue or a refinement-capability issue, the repository includes a feasibility sweep over transfer time, thrust limit, segment count, and least-squares budget. It also includes direct multiple-shooting diagnostics with nominal and selected-outage branch states and controls, RK4 segment defects, linear state-node initialization, optional nominal-first homotopy, and row-level settings fingerprints for resumed mixed runs. The newest bounded-projected variant applies the acceleration-ball projection inside the residual unpacking path, so defect propagation, terminal residuals, control penalties, smoothness penalties, evaluation, and reported controls all use the same norm-bounded accelerations. A final targeted catalog feasibility attempt uses stronger multistart branch recovery at $T = 4.0$ TU, $a_{\max} = 0.3$, $N = 14$, and a 250-evaluation limit. These tests are not used to claim a best trajectory; they are used to locate the boundary of the present implementation.

5.5. Teacher-controlled feasible benchmark

The teacher-controlled benchmark generates the target by forward propagation from s_0 under a disclosed deterministic bounded low-thrust policy. The active schedule is

$$0000011110101010, \tag{16}$$

with active fraction 0.4375, transfer time $T = 4.0$ TU, $N = 16$, $a_{\max} = 0.065$, teacher maximum control norm 0.039, and teacher fuel 0.0625147492 in normalized acceleration-time units. The generated target is

$$\begin{aligned} s_f = [&0.9072480273, -0.0480603610, -0.1216659997, \\ &-0.1816901396, -0.0351710714, 0.1330325296]. \end{aligned} \tag{17}$$

The teacher reproduces its own target with zero propagation error under the same integrator, while its worst one- or two-segment missed-thrust outage without recovery is 0.6049 under the configured error norm. This benchmark is therefore feasible by construction but not trivial under outages. It is included to test whether the initialization/refinement architecture can recover a known feasible low-thrust transfer without hiding the construction. The oracle diagnostic initializes the refinement directly with the teacher continuous controls and is labeled separately from the normal method comparison.

Table 1: Catalog-derived pilot results. Values are medians across the generated pilot seeds. This benchmark did not meet the robust refinement thresholds.

Method	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
random	1.8309	2.5333	34.0442	0.3571	120.0000	0.0000	120.0000	0.0000
cross_entropy	0.9023	0.9031	9.9415	0.5000	120.0000	0.0000	120.0000	0.0000
genetic	2.5371	2.6132	34.8813	0.5000	132.0000	0.0000	132.0000	0.0000
true_sa	2.3540	2.4855	34.2025	0.6429	120.0000	0.0000	120.0000	0.0000
surrogate_qubo_sa	2.6497	2.7770	33.8472	0.5714	48.0000	72.0000	120.0000	0.0000
qaoa_statevector	2.4262	2.5573	33.8785	0.6429	48.0000	72.0000	120.0000	0.0000
all_windows_continuous	1.6383	2.2746	23.7862	1.0000	31.0000	0.0000	31.0000	0.0000

5.6. Metrics

The primary metrics are refined nominal terminal error, refined selected missed-thrust worst error, all-mask diagnostic worst error, schedule active fraction, nominal and recovery fuel, total true trajectory evaluations, and refinement success rate. When schedule repair is enabled, the tables separately report the solver's best schedule, the schedule selected for refinement, the repaired refined schedule, and their active fractions, so that a sparse candidate cannot be confused with a dense refined envelope. The $N = 8$ phase-shift experiments use five deterministic seeds. The $N = 12$ cardinality-prior experiment uses ten deterministic seeds, 96 shared QUBO-training evaluations, and 24 post-QUBO candidate evaluations for QUBO/QAOA methods; equal-budget classical methods receive 120 true objective evaluations, matching the QUBO methods after shared training is charged. The teacher-controlled experiment uses three deterministic seeds. Unless otherwise stated, the thresholds are 0.09 for nominal success and 0.17 for selected robust recovery success.

6. Results

6.1. Catalog-derived benchmark remains unresolved

Table 1 reports the catalog-derived pilot. QUBO simulated annealing and QAOA produce competitive pre-refinement schedules in this small run, but after fair accounting they require shared QUBO-training evaluations, and no method satisfies the configured post-refinement robust success thresholds. This is a negative result: a quantum-ready schedule sampler is not enough when the continuous recovery optimizer cannot turn the schedule into a feasible robust trajectory.

The QUBO diagnostics in Table 2 also show overfitting risk. With a full quadratic model and a small training set, the training fit is much stronger than the validation behavior. This motivates reporting QUBO fit quality as a first-order result rather than assuming that a learned quadratic model preserves the true trajectory ranking.

Table 2: Catalog-derived QUBO surrogate fit diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0002	4.3189	1.0000	0.5063	54	18	72	72	True
23	0.0001	0.0002	4.0060	1.0000	0.7190	54	18	72	72	True
37	0.0001	0.0002	5.9296	1.0000	0.0352	54	18	72	72	True

Table 3: Feasibility sweep stress test for the catalog-derived benchmark. None of the completed cases met both nominal and selected robust thresholds.

Transfer time	amax	Segments	Max nfev	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Meets thresholds	Best start	nfev	Runtime (s)
4.0000	0.2000	14	120	0.2008	0.2310	2.8975	False	feedback	251	1635.4279
4.0000	0.3000	14	150	0.1229	0.2783	7.5962	False	feedback	177	1102.4643
4.0000	0.3000	18	150	0.2423	0.2823	5.0379	False	feedback	39	351.0886
5.0000	0.3000	14	150	0.5810	0.6327	57.6216	False	zero	30	88.5080
5.0000	0.1500	14	120	0.6248	0.6785	53.6764	False	zero	138	997.2723
4.0000	0.2500	14	150	1.1558	1.2770	60.9139	False	feedback	81	324.7229
4.0000	0.1500	14	120	0.7389	1.6931	37.8087	False	zero	32	122.3204
3.0000	0.0650	14	120	1.0202	2.5174	7.1874	False	zero	61	326.9638

6.2. Feasibility stress tests expose the refinement boundary

Table 3 shows the best completed points from the feasibility sweep. The best nominal error found in the sweep is 0.1229–0.2008 depending on the point, but selected outage errors remain above the 0.17 robust threshold and all-mask diagnostics remain much larger. Increasing thrust or transfer time does not monotonically fix the problem because the nonlinear dynamics and local least-squares convergence interact with the initialization.

The direct multiple-shooting diagnostics in Table 4 extend this stress test to six completed catalog-derived rows. The best nominal-only row, $T = 5.0$ TU and $a_{\max} = 0.5$, reaches nominal error 0.5154 but has an all-mask diagnostic worst error of 55.2845. The best selected-outage row, $T = 4.0$ TU and $a_{\max} = 0.5$, has nominal error 1.5355 and selected recovery worst error 1.2268. A nominal-first homotopy row reduces the all-mask diagnostic worst error to 5.5503 but worsens the selected recovery worst error to 4.3964. None of the rows satisfies the 0.09 nominal and 0.17 selected-recovery thresholds. This does not prove that multiple shooting is unsuitable; it shows that the present sparse least-squares implementation and budget are insufficient for the hard catalog-derived case.

Table 5 reports the stronger targeted catalog attempt. Despite a larger thrust bound, multistart initialization, bang-bang feedback starts, and a 250-evaluation limit, the best completed case misses both thresholds. The optimizer selected a low-control solution with nominal error 0.3055 and selected recovery worst error 1.3513. A $N = 18$ variant was terminated after more than nine minutes without a completed row. This negative result strengthens the interpretation that the present continuous backend is not a mature catalog-transfer optimizer.

Table 4: Catalog-derived direct multiple-shooting diagnostics. Rows with zero selected outages are nominal-only diagnostics; all-outage diagnostic errors are still reported.

Transfer time	amax	Segments	Selected outages	Initialization	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Fuel	nfev	Runtime (s)	Meets thresholds
5.0000	0.5000	14	0	linear	0.5154	0.5154	55.2845	2.5000	100	43.4353	False
4.0000	0.5000	14	3	linear	1.5355	1.2268	25.1426	2.0000	100	118.9605	False
4.0000	0.3000	14	3	linear	1.5981	1.9548	20.1304	1.2000	14	22.9601	False
4.0000	0.3000	14	3	linear	1.6273	4.3964	5.5503	1.2000	74	172.0316	False
5.0000	0.5000	14	3	linear	6.1753	5.7643	9.0650	2.5000	30	68.2684	False
6.0000	0.5000	14	0	linear	7.4830	7.4830	119.2412	3.0000	100	49.9072	False

Table 5: Targeted catalog-derived feasibility attempt with stronger multistart branch recovery. Thresholds were not relaxed.

Transfer time	amax	Segments	Max nfev	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Meets thresholds	Best start	nfev	Runtime (s)
4.0000	0.3000	14	250	0.3055	1.3513	5.4449	False	zero	92	389.4414

6.3. Bounded projected multiple shooting solves limited non-teacher phase-shift cases

Table 6 reports the compact bounded-projected multiple-shooting benchmark on the catalog halo phase-shift target. The positive row uses $N = 12$, $a_{\max} = 0.2$, one selected outage branch, linear node initialization, and a 40-evaluation cap. It reaches nominal error 0.0479, selected recovery worst error 0.0619, and all-mask diagnostic worst error 0.0752, all below the configured thresholds. The reported maximum acceleration norm is 0.2, and the largest bound violation is 5.6×10^{-17}, i.e., numerical roundoff.

This result is a backend result rather than a quantum-sampling result. The selected outage branch, index 6 in the saved metadata, is optimized with branch recovery. The all-mask diagnostic evaluates every one-segment outage; unselected outages use the nominal controls with the outage mask applied, not separately optimized recovery branches. The diagnostic is therefore stricter than selected-branch success in one sense, because every unselected one-segment outage remains below threshold without its own recovery solve, but it is not the same as optimizing all outage branches. The rows selecting all one-segment outage branches at $N = 8$ and $N = 12$ fail the robust threshold under their smaller completed budgets, with selected/all-mask worst errors 0.3378 and 0.3904, respectively. Thus the bounded projected backend fixes a real control-bound inconsistency and provides a limited non-teacher positive result, but it does not close the hard catalog-transfer gap.

Table 7 extends this backend diagnostic across phase times 0.2, 0.3, and 0.4 with row-level settings fingerprints, configuration hashes, and source-state identifiers so that resumed rows cannot be silently reused after a settings change. Two of the three selected-branch rows meet the thresholds. The phase-time 0.4 row is the strongest completed case, with nominal error

Table 6: Bounded-projected multiple-shooting diagnostics on the non-teacher catalog halo phase-shift target. Controls are projected to $\|u_i\| \leq a_{\max}$ inside the residual and evaluation paths.

Transfer time	amax	Segments	Selected outages	Initialization	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Fuel	Max ||u||	Bound violation	nfev	Runtime (s)	Meets thresholds
0.5000	0.2000	12	1	linear	0.0479	0.0619	0.0752	0.1000	0.2000	0.0000	24	18.0024	True
0.5000	0.2000	8	8	linear	0.0414	0.3378	0.3378	0.1000	0.2000	0.0000	30	73.0455	False
0.5000	0.2000	12	12	linear	0.0396	0.3904	0.3904	0.1000	0.2000	0.0000	20	141.5146	False

Table 7: Bounded-projected multiple-shooting phase-suite diagnostics on non-teacher catalog halo phase-shift targets. The selected-branch rows optimize one outage branch; the all-outage-selected row optimizes all one-segment outage branches and remains infeasible.

Case	Phase time	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ||u||	Bound violation	nfev	Runtime (s)	Meets thresholds
all_outage_selected_diagnostic	0.3000	0.5000	0.2000	12	12	0.0396	0.3581	0.3581	0.2000	0.0000	40	366.6997	False
selected_branch	0.2000	0.5000	0.2000	12	1	0.1939	0.2168	0.2288	0.2000	0.0000	22	12.6848	False
selected_branch	0.3000	0.5000	0.2000	12	1	0.0479	0.0619	0.0752	0.2000	0.0000	24	17.5261	True
selected_branch	0.4000	0.5000	0.2000	12	1	0.0260	0.0083	0.0274	0.2000	0.0000	40	40.3375	True

0.0260, selected recovery worst error 0.0083, and all-mask diagnostic worst error 0.0274. The phase-time 0.2 row fails both thresholds, and an all-outage-selected diagnostic at phase time 0.3 fails with selected/all-mask worst error 0.3581. The broader suite therefore strengthens the continuous-backend evidence while preserving the main limitation: selected-branch feasibility over one outage does not imply successful simultaneous optimization over every outage branch.

Table 8 adds a separate bounded projected trapezoidal direct-collocation baseline on the same phase-shift suite. The direct-collocation variables include nominal state nodes, nominal controls, selected outage branch nodes, and selected branch recovery controls; the residual enforces trapezoidal dynamics defects and terminal conditions while projecting every reported and evaluated control vector to $\|u_i\| \leq a_{\max}$. This baseline succeeds only for phase time 0.4, with nominal error 0.0357, selected recovery worst error 0.0191, all-mask diagnostic worst error 0.0602, and maximum control norm 0.1483. It fails at phase times 0.2 and 0.3 under the compact 30-evaluation budget. The result addresses the missing-continuous-baseline concern in a limited way: a recognizable direct transcription agrees that the easier phase-time 0.4 case is solvable, but the compact implementation is not a mature production collocation or continuation workflow and does not solve the harder phase-shift rows.

6.4. Non-teacher phase-shift benchmark is solved only by the continuous all-windows baseline

Table 9 reports the catalog halo phase-shift benchmark. This case removes the teacher-generation objection: the target is a zero-thrust CR3BP

Table 8: Compact bounded projected trapezoidal direct-collocation baseline on the non-teacher catalog halo phase-shift suite. Rows optimize one selected outage branch and report all-mask diagnostics separately.

Case	Phase time	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ‖u‖	Bound violation	nfev	Runtime (s)	Meets thresholds
selected_branch	0.2000	0.5000	0.2000	12	1	0.4512	0.4794	0.4805	0.2000	0.0000	30	43.4778	False
selected_branch	0.3000	0.5000	0.2000	12	1	0.3714	0.3774	0.3969	0.2000	0.0000	30	48.8181	False
selected_branch	0.4000	0.5000	0.2000	12	1	0.0357	0.0191	0.0602	0.1483	0.0000	11	18.9927	True

Table 9: Catalog halo phase-shift benchmark. The target is generated by ballistic CR3BP phase propagation of the JPL source state, not by a low-thrust teacher.

Benchmark	Method	Comparison group	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	cross_entropy	method_comparison	0.2663	0.3042	0.3042	0.3750	56.0000	0.0000	56.0000	0.0000
catalog_halo_phase_shift	genetic	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	true_sa	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	surrogate_qubo_sa	method_comparison	0.2663	0.3042	0.3042	0.3750	20.0000	32.0000	52.0000	0.0000
catalog_halo_phase_shift	qaoa_statevector	method_comparison	0.2242	0.2625	0.2650	0.5000	20.0000	32.0000	52.0000	0.0000
catalog_halo_phase_shift	all_windows_continuous	method_comparison	0.0418	0.0750	0.0887	1.0000	30.0000	0.0000	30.0000	1.0000

phase point of the JPL L2 southern halo source state, and the saved metadata records `teacher_generated=false`. The all-windows continuous baseline succeeds in all five seeds, with median nominal error 0.0418, median selected recovery worst error 0.0750, and median all-mask diagnostic worst error 0.0887. The result also exposes a different limitation: none of the tested binary schedule samplers finds a schedule that refines to the thresholds in this five-seed run. QAOA gives the best median among the discrete initializers, but its median selected recovery error remains 0.2625, above the 0.17 threshold.

The phase-shift QUBO surrogate is not the limiting factor in the same way as in the teacher benchmark. Across five seeds, validation R^2 ranges from 0.321 to 0.955 and validation Spearman correlation ranges from 0.310 to 1.000. The remaining failure is therefore more structural: the robust binary objective and the tested samplers prefer sparse late-thrust schedules such as `00000111` or `00001111`, while the successful baseline uses all thrust windows and lets the continuous optimizer shape the acceleration. In this case, the evidence favors continuous-control freedom over sparse binary schedule selection.

6.5. Dense schedule repair exposes the sparsity bottleneck

Table 10 reports a deterministic schedule-repair ablation on the same non-teacher phase-shift benchmark. The repair is `prefix_to_last_active`: before continuous refinement, every window through the last active solver window is made available for thrust, while later zeros are preserved. This is a heuristic dense-availability envelope, not an oracle continuous control and not a quantum routine.

Table 10: Catalog halo phase-shift benchmark with deterministic dense-schedule repair. The table exposes the solver schedule, refinement candidate schedule, repaired refined schedule, and active fractions to show that all repaired methods collapse to the same all-windows refinement envelope.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
catalog_halo_phase_shift	cross_entropy	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	56.0000	0.0000	56.0000	1.0000
catalog_halo_phase_shift	genetic	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	00000111	00000111	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
catalog_halo_phase_shift	true_sa	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	00000111	00000111	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	20.0000	32.0000	52.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.5000	0.5000	1.0000	20.0000	32.0000	52.0000	1.0000
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	11111111	11111111	11111111	0.0418	0.0750	0.0887	1.0000	1.0000	1.0000	30.0000	0.0000	30.0000	1.0000

Table 11: Catalog halo phase-shift benchmark with a 12-segment duty-cycle prior targeting one coast window. QUBO/QAOA totals include 96 shared QUBO-training evaluations and 24 post-QUBO true candidate evaluations.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Active target penalty	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0954	0.1214	0.1313	0.8333	0.0833	0.8333	0.8333	120.0000	0.0000	120.0000	0.2000
catalog_halo_phase_shift	cross_entropy	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	genetic	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0675	0.0926	0.1020	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	true_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	24.0000	96.0000	120.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0942	0.1201	0.1292	0.8333	0.0833	0.8333	0.8333	24.0000	96.0000	120.0000	0.4000
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	111111111111	111111111111	111111111111	0.0386	0.0601	0.0690	1.0000	0.0833	1.0000	1.0000	19.0000	0.0000	19.0000	1.0000

With repair enabled, every tested initializer succeeds in 5/5 seeds and attains the same median refined errors as the all-windows continuous baseline: nominal error 0.0418, selected recovery worst error 0.0750, and all-mask diagnostic worst error 0.0887. The provenance columns are the important result. Repaired methods enter refinement with sparse candidate active fractions, typically 0.375 and 0.500, but every method refines the same dense schedule `11111111` with refined active fraction 1.0. Therefore, this ablation does not rank the samplers and does not establish a quantum advantage. It shows that the current binary objective can place thrust late enough for a deterministic dense repair to recover a useful all-windows envelope, while the unrepaired binary encoding is too sparse for the continuous backend.

6.6. Duty-cycle prior yields a positive non-teacher discrete-initialization result

Table 11 reports the $N = 12$ phase-shift benchmark with the active-fraction prior J_ρ set to $\rho = 11/12$. This is the first non-teacher case in the paper where the discrete layer contributes a useful refinement structure rather than only diagnosing failure. Cross-entropy, genetic search, true-objective simulated annealing, and surrogate-QUBO simulated annealing each succeed in 10/10 seeds and select one-coast schedules with median refined active fraction 0.9167. Random search succeeds in 2/10 seeds and QAOA succeeds in 4/10 seeds under the same total true-evaluation budget.

The result is a tradeoff, not a dominance claim. The all-windows continuous baseline also succeeds in 10/10 seeds and has lower terminal errors: median nominal error 0.0386 and selected recovery worst error 0.0601. The one-coast methods have higher but feasible errors, with surrogate-QUBO

Table 12: QAOA depth and angle-optimization ablation on the $N = 12$ duty-cycle-prior phase-shift benchmark. All rows use 96 shared QUBO-training trajectory evaluations and 24 true post-QUBO candidate evaluations per seed.

method	runs	refinement_success_rate	refined_nominal_error_median	refined_selected_worst_error_median	refined_nominal_fuel_median	solver_true_evaluations_median	shared_qubo_training_evaluations_median	runtime_seconds_median
surrogate_qubo_sa	10	1.0000	0.0682	0.0933	0.0917	24.0000	96.0000	0.3912
exact_qubo_top24	10	0.9000	0.0690	0.0941	0.0917	24.0000	96.0000	0.3957
qaoa_random_p1	10	0.3000	0.0948	0.1203	0.0833	24.0000	96.0000	0.4140
qaoa_optimized_p1	10	0.6000	0.0690	0.0941	0.0917	24.0000	96.0000	0.4098
qaoa_optimized_p2	10	0.9000	0.0690	0.0941	0.0917	24.0000	96.0000	0.4964

simulated annealing at median nominal error 0.0690 and selected recovery worst error 0.0941. Their apparent benefit against the default all-windows setting is lower nominal fuel: 0.0917 versus 0.1000, an 8.33% reduction in this normalized fixed-time metric. The QUBO surrogate is well aligned with the modified objective in this benchmark, with mean validation $R^2 = 0.969$, mean validation Spearman correlation 0.962, mean pairwise order accuracy 0.935, and mean top-k recall 0.86 across ten seeds. Thus, the evidence supports a narrow claim: a duty-cycle-aware QUBO objective can recover a refinable high-duty schedule class on a non-teacher cislunar benchmark, but strong classical heuristics match it and the initially tested shallow random-angle QAOA sampler does not.

6.7. Optimized QAOA angles improve the QUBO sampler but do not dominate

Table 12 isolates the QUBO-sampling step on the same $N = 12$ duty-cycle-prior benchmark. All methods use the same ten seeds, the same fitted QUBO training sets, and the same accounting: 96 shared trajectory evaluations to train the QUBO and 24 true post-QUBO candidate evaluations per method and seed. The random-angle $p = 1$ QAOA sampler succeeds in 3/10 seeds in this focused run. Optimizing QAOA angles by minimizing expected surrogate QUBO energy improves selected-branch refinement success to 6/10 for $p = 1$ and 9/10 for $p = 2$, and reduces the median selected worst error from 0.1203 to 0.0941. The optimized $p = 2$ result matches exact evaluation of the 24 best QUBO bitstrings among all schedules, but surrogate-QUBO simulated annealing still succeeds in 10/10 seeds with comparable median error. Therefore the ablation supports a useful but limited quantum-ready result: QAOA angle optimization matters, but this statevector QAOA configuration does not outperform a strong classical QUBO sampler.

6.8. Cardinality ablation and fuel-error Pareto check

Table 13 places the duty-cycle result in context by exhaustively refining high-duty schedule classes. All 12 one-coast schedules are feasible, and 41

Table 13: Exhaustive high-duty cardinality ablation for the $N = 12$ phase-shift benchmark.

k_active	schedules_refined	success_count	all_mask_success_count	best_nominal_error	median_nominal_error	best_selected_worst_error	median_selected_worst_error	best_all_mask_worst_error	median_all_mask_worst_error	best_nominal_fuel	median_nominal_fuel	best_schedule_by_selected_worst	min_fuel_feasible_schedule
10	66	41	41	0.0798	0.0874	0.1056	0.1132	0.1146	0.1230	0.0833	0.0833	111111111100	111111011110
11	12	12	12	0.0588	0.0617	0.0833	0.0864	0.0923	0.0956	0.0917	0.0917	111111111110	111111111011
12	1	1	1	0.0386	0.0386	0.0601	0.0601	0.0690	0.0690	0.1000	0.1000	111111111111	111111111111

Table 14: All-windows fuel-regularization sweep for the $N = 12$ phase-shift benchmark.

fuel_residual_weight	refinement_success	all_mask_success	refined_nominal_error	refined_selected_worst_error	refined_all_mask_worst_error	refined_nominal_fuel	refinement_nfev
0.0180	True	True	0.0386	0.0601	0.0690	0.1000	19
0.0300	True	True	0.0468	0.0711	0.0791	0.0962	18
0.0500	True	True	0.0773	0.1100	0.1122	0.0846	15
0.0800	False	False	0.1204	0.1491	0.1564	0.0700	10
0.1200	False	False	0.1891	0.2128	0.2183	0.0504	7
0.2000	False	False	0.2755	0.2892	0.2915	0.0268	7

of 66 two-coast schedules are feasible. This means the duty-cycle prior is not discovering a rare one-coast structure; it is steering the search into a broad feasible high-duty region. Within the one-coast class, surrogate-QUBO simulated annealing selects feasible schedules in 10/10 runs, but none of those schedules lies on the one-coast Pareto frontier used by the ablation, and its median selected-worst gap to the best one-coast schedule is 0.0108.

Table 14 then checks whether the one-coast fuel reduction survives a stronger continuous all-windows baseline. It does not. Increasing the all-windows fuel residual weight to 0.05 keeps the trajectory feasible, with nominal error 0.0773, selected recovery worst error 0.1100, all-mask worst error 0.1122, and nominal fuel 0.0846, lower than the one-coast median fuel 0.0917. The Pareto interpretation is therefore that binary duty-cycle control finds a feasible high-duty schedule class, while continuous fuel regularization inside all available windows is currently more effective at trading terminal error for fuel in this benchmark.

6.9. Teacher-controlled benchmark demonstrates selected-branch recovery

Table 15 reports the teacher-controlled benchmark. Random search succeeds in 2/3 seeds. Cross-entropy, surrogate QUBO simulated annealing, and QAOA each succeed in 1/3 seeds. Genetic search, true-objective simulated annealing, the all-windows continuous baseline, and the teacher-seeded binary schedule do not meet the selected-branch thresholds in this three-seed run. The teacher-controls oracle diagnostic succeeds in 3/3 seeds, with median nominal error 0.0067 and selected recovery worst error 0.0080, but it is not a valid competing initializer because it receives the hidden continuous controls used to generate the benchmark target.

The result has two interpretations. The positive interpretation is that the proposed architecture can produce selected-branch robust refinements on a feasible low-thrust transfer, including through QUBO and QAOA initializers.

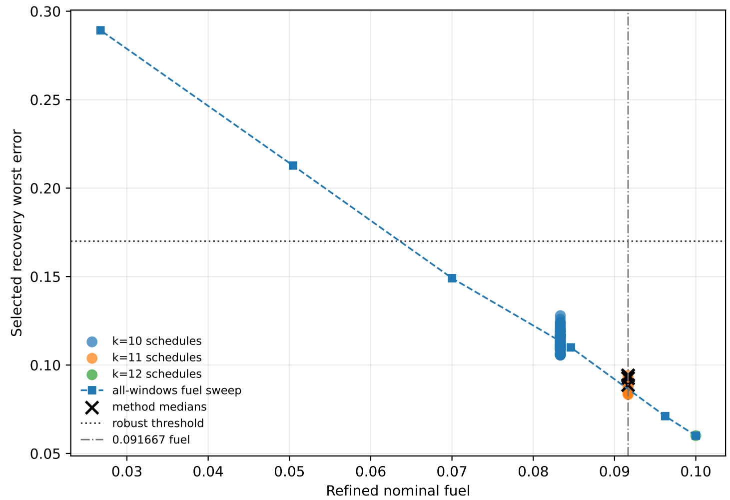

Figure 1: Fuel-error Pareto view for the duty-cycle-prior ablation. Lower fuel is achievable with all-windows fuel regularization while retaining feasibility, so the one-coast result should be read as a schedule-structure feasibility result rather than a fuel-optimality result.

Table 15: Teacher-controlled benchmark results. Values are medians across three seeds. QUBO and QAOA totals include the shared QUBO-training trajectory evaluations.

Benchmark	Method	Comparison group	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
teacher_controlled	random	method_comparison	0.0115	0.0137	1.1009	0.3750	52.0000	0.0000	52.0000	0.6667
teacher_controlled	cross_entropy	method_comparison	0.0381	0.2626	0.7695	0.3750	56.0000	0.0000	56.0000	0.3333
teacher_controlled	genetic	method_comparison	0.0213	0.5602	0.8393	0.3125	52.0000	0.0000	52.0000	0.0000
teacher_controlled	true_sa	method_comparison	0.6349	0.9638	16.4017	0.6250	52.0000	0.0000	52.0000	0.0000
teacher_controlled	surrogate_qubo_sa	method_comparison	0.0263	0.2162	0.8145	0.3750	20.0000	32.0000	52.0000	0.3333
teacher_controlled	qaoa_statevector	method_comparison	0.2495	0.2496	1.0213	0.3125	20.0000	32.0000	52.0000	0.3333
teacher_controlled	all_windows_continuous	method_comparison	6.3495	8.4578	17.8387	1.0000	3.0000	0.0000	3.0000	0.0000
teacher_controlled	teacher_seeded_schedule	method_comparison	1.2183	0.9547	1.7652	0.4375	30.0000	0.0000	30.0000	0.0000
teacher_controlled	teacher_controls_oracle_diagnostic	oracle_diagnostic	0.0067	0.0080	1.1910	0.4375	7.0000	0.0000	7.0000	1.0000

The limiting interpretation is more important for publication: random search currently has the highest non-oracle success rate, all-mask diagnostic errors remain much larger than selected-branch errors, and the oracle result shows that continuous-control initialization can dominate the outcome. Therefore the present data support feasibility of the pipeline and diagnose a bottleneck, not superiority of quantum-ready sampling.

The seed counts are still small. Wilson 95% intervals are wide: the phase-shift all-windows success rate of $5/5$ has interval approximately $[0.57, 1.00]$, the unrepaired phase-shift discrete-method rates of $0/5$ have intervals approximately $[0, 0.43]$, and the repaired phase-shift method rates of $5/5$ again have intervals approximately $[0.57, 1.00]$ but are method-collapsed by the deterministic repair. In the cardinality-prior case, $10/10$ rates have intervals approximately $[0.72, 1.00]$, random search at $2/10$ has interval approximately

Table 16: Teacher-controlled recovery fuel and refinement effort. Fuel is reported in normalized acceleration-time units.

Method	Nominal fuel	Mean recovery fuel	Max recovery fuel	Refinement nfev	Recovery success
random	0.0579	0.0579	0.0579	24.0000	0.6667
cross_entropy	0.0892	0.0813	0.0813	30.0000	0.3333
genetic	0.0679	0.0650	0.0650	30.0000	0.0000
true_sa	0.1592	0.1355	0.1438	16.0000	0.0000
surrogate_qubo_sa	0.0733	0.0650	0.0650	27.0000	0.3333
qaoa_statevector	0.0488	0.0488	0.0488	30.0000	0.3333
all_windows_continuous	0.2595	0.2194	0.2284	3.0000	0.0000
teacher_seeded_schedule	0.1134	0.0793	0.0849	30.0000	0.0000
teacher_controls_oracle_diagnostic	0.0281	0.0272	0.0283	7.0000	1.0000

Table 17: Teacher-controlled QUBO surrogate fit diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	val_spearman	val_pairwise_order_accuracy	val_top_k	val_top_k_recall	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0001	9.4732	1.0000	0.0443	0.7143	0.7857	2	0.5000	24	8	32	32	True
23	0.0001	0.0002	13.1807	1.0000	-1.7931	0.6190	0.7857	2	1.0000	24	8	32	32	True
37	0.0001	0.0001	7.2534	1.0000	0.6466	0.5714	0.6786	2	1.0000	24	8	32	32	True

$[0.06, 0.51]$, and QAOA at 4/10 has interval approximately $[0.17, 0.69]$. The teacher random-search rate of 2/3 has interval approximately $[0.21, 0.94]$, and the teacher QUBO/QAOA rates of 1/3 have intervals approximately $[0.06, 0.79]$. These intervals are reported to prevent overinterpretation; they do not support a statistically significant ranking of all stochastic initializers.

Table 16 reports nominal and recovery fuel. QAOA has the lowest median fuel among methods with nonzero recovery success, but its selected robust error and success rate are worse than random search. Fuel therefore cannot be evaluated independently of feasibility.

6.10. Surrogate quality remains a bottleneck

Table 17 shows teacher-controlled QUBO diagnostics. Validation R^2 varies from negative to moderately positive across only three seeds. Rank diagnostics are more favorable but still not decisive: validation Spearman correlation is 0.57–0.71, pairwise order accuracy is 0.68–0.79, and top-2 recall is 0.5–1.0 on only eight validation samples. This is too weak for a final surrogate-generalization claim and likely contributes to the lack of QUBO/QAOA dominance. A submission-ready version needs larger training sets, cross-validated regularization, ranking-oriented validation, or a structured surrogate with trajectory-informed features.

7. Discussion

The experiments support ten conclusions. First, robust low-thrust initialization can be represented as a binary scheduling problem with trajectory-

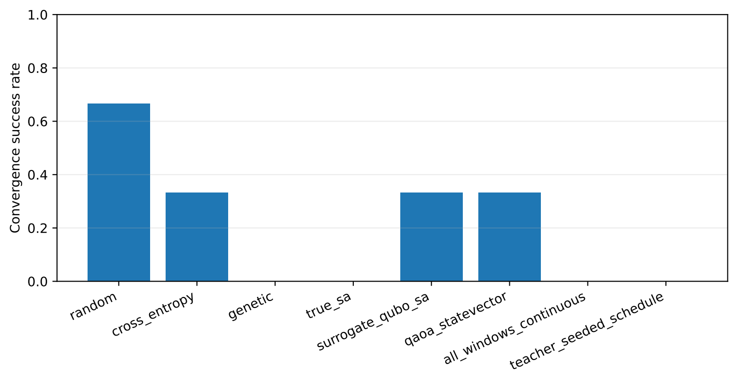

Figure 2: Teacher-controlled selected-branch refinement success rates. The result demonstrates nonzero recovery success but not quantum advantage.

evaluated missed-thrust penalties and then approximated by a QUBO. Second, the phase-shift benchmark shows that the present branch-recovery backend can solve a non-teacher catalog/dynamics-derived case under the configured thresholds, including all one-segment outage masks. Third, enforcing the acceleration bound inside the multiple-shooting residual removes an important implementation inconsistency and produces limited positive non-teacher backend results, although all-branch optimization remains hard. Fourth, a separate bounded trapezoidal direct-collocation baseline succeeds on the easiest phase-suite row and fails on the two harder rows, supporting the interpretation that backend capability is strongly case-dependent. Fifth, that same phase-shift benchmark shows that the tested binary schedule samplers can fail even when the continuous all-windows baseline succeeds; the discrete layer is currently too sparse and too dependent on the steering-law objective. Sixth, deterministic dense-schedule repair can make the phase-shift benchmark refinable for all samplers, but only by collapsing them to the same all-windows refined envelope, so the repair is diagnostic rather than discriminating. Seventh, adding a duty-cycle prior creates a positive but limited non-teacher result: the QUBO-ready objective can select one-coast schedules that remain robustly refinable, although exhaustive cardinality and all-windows fuel-regularization ablations show that this is a broad feasible schedule class rather than a superior Pareto frontier. Eighth, optimized-angle QAOA improves markedly over random-angle QAOA and $p = 2$ matches exact top-24 QUBO candidate evaluation in the focused abla-

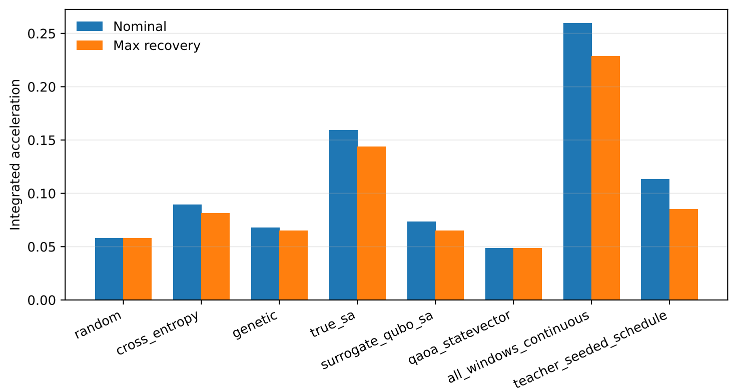

Figure 3: Teacher-controlled nominal and recovery fuel summaries. Fuel differences must be interpreted together with feasibility and missed-thrust error.

tion, but it still does not outperform surrogate-QUBO simulated annealing. Ninth, selected-branch robust refinement is possible in a disclosed teacher-controlled cislunar benchmark, but the current success rates are low and do not favor QUBO/QAOA over all classical baselines. Tenth, the hard NRHO-like to DRO-phase catalog-derived case exposes the gap between this research scaffold and mature cislunar low-thrust design: even stronger targeted multi-start and row-truthful direct multiple-shooting diagnostics do not reach the thresholds.

The teacher diagnostics are particularly revealing. Even when the binary schedule that generated the teacher target is supplied, the branch-recovery optimizer does not recover a successful selected-outage solution under the current feedback initialization. The oracle continuous-control diagnostic then succeeds on every seed. This pair of results indicates that schedule structure alone is insufficient and that the present feedback control initializer can place the least-squares refinement outside the useful basin. Continuous thrust initialization, branch selection, least-squares scaling, and recovery parameterization all need improvement before the discrete initializer can be judged fairly.

The phase-shift diagnostics point to a complementary failure mode. There, the continuous all-windows baseline succeeds without teacher controls, and the bounded-projected multiple-shooting rows confirm that a consistent norm-bounded residual can solve selected-branch versions while keeping all one-segment outage diagnostics below threshold in the

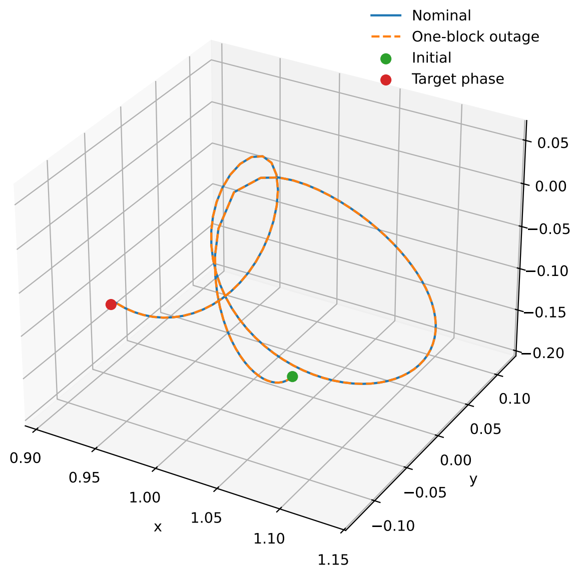

Figure 4: Example trajectory from the teacher-controlled benchmark. The case is a normalized CR3BP algorithm benchmark, not a flight-ready trajectory.

successful rows. The trapezoidal direct-collocation baseline independently solves the phase-time 0.4 row but fails at 0.2 and 0.3, so it narrows but does not eliminate the continuous-backend concern. The unrepaired binary samplers still do not select enough useful thrust windows. The dense-repair ablation confirms the interpretation: when sparse late-window candidates are expanded into the all-windows envelope, the same continuous backend succeeds, but the repair removes any meaningful method ranking. The cardinality-prior experiment then shows one way to make the binary layer useful: explicitly encode a desired duty cycle so that the sampler preserves enough low-thrust availability while still removing one window. The optimized-QAOA ablation improves the quantum-ready sampler itself: moving from random $p = 1$ angles to optimized $p = 2$ angles raises success from $3/10$ to $9/10$ under the same post-QUBO candidate budget. However, the exhaustive ablation shows that all one-coast schedules and many two-coast schedules are feasible, the all-windows fuel sweep shows that continuous fuel regularization is a stronger Pareto mechanism in this case, and QUBO simulated annealing remains stronger than the tested QAOA. This suggests that the present binary encoding is too coarse without

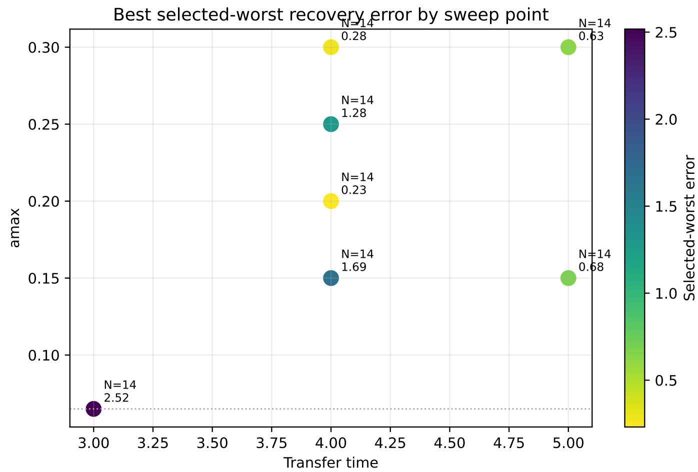

Figure 5: Feasibility sweep heatmap for the catalog-derived benchmark. The completed cases remain outside the configured robust success thresholds.

additional trajectory-shaping information. A stronger formulation may need multi-level thrust availability, direction classes, learned repair policies with held-out validation, or a two-stage objective that preserves dense low-thrust arcs while optimizing fuel in continuous control space.

For a Q1 aerospace journal, the present package is therefore not yet a quantum-performance paper. It is closer to a methods-and-evidence paper that documents a realistic hybrid architecture, non-teacher solvable control cases, a compact direct-collocation comparison, one duty-cycle-aware discrete-initialization result, an optimized-QAOA improvement that remains below the best classical QUBO sampler, and several failure modes with reproducible provenance. To become a strong performance paper, the next version must show statistically meaningful improvement over strong classical initializers under equal total trajectory-evaluation budgets, include more seeds and confidence intervals, strengthen QUBO/QAOA studies beyond shallow simulated sampling, and use a more capable continuous optimal-control backend such as Hermite–Simpson collocation, continuation, or sequential convex programming.

8. Limitations

The main limitations are as follows. The phase-shift benchmarks are non-teacher and not do-nothing passes, but they are still short normalized CR3BP retargeting cases and are easier than the NRHO-like to DRO-phase transfer. The bounded-projected multiple-shooting and direct-collocation results are continuous-backend improvements, not binary or quantum-sampling results, and their positive selected-branch rows optimize one selected outage branch while reporting unselected outages as diagnostics using masked nominal controls. The direct-collocation baseline is trapezoidal and compact, not Hermite–Simpson and not a production continuation workflow; it succeeds on only one of the three phase-suite rows. The all-outage-selected phase-suite row remains infeasible. The unrepaired $N = 8$ success comes from the all-windows continuous baseline, not from the quantum-ready schedule samplers. The dense-repair ablation succeeds by converting sparse candidates to `11111111`, so it should be read as an encoding diagnostic rather than a competitive result. The $N = 12$ duty-cycle-prior result is positive but narrow: it demonstrates refinable high-duty schedules, not lower terminal error than all-windows, not a Pareto fuel advantage over all-windows with stronger fuel regularization, and not superiority over strong classical CEM, genetic search, or true-objective simulated annealing. Optimized-angle QAOA is tested only in statevector simulation with $p \leq 2$ and a reduced optimizer budget of one restart and ten iterations; it improves over random-angle QAOA but does not outperform surrogate-QUBO simulated annealing. The teacher-controlled benchmark is feasible by construction and should not be confused with an operational transfer. The hard catalog-derived benchmark remains unresolved under the present thresholds, including under the stronger targeted multi-start and direct multiple-shooting diagnostic attempts. Ten cardinality-prior seeds, five $N = 8$ phase-shift seeds, and three teacher-controlled seeds are still insufficient for strong statistical claims. The QUBO surrogate has mixed validation performance across benchmarks. Branch recovery optimizes only selected outage masks and is evaluated separately against all masks, except in the successful phase-shift cases where all one-segment outages also remain below threshold for the refined schedules. The all-windows, multiple-shooting, and compact direct-collocation baselines are implementation baselines, not mature production trajectory optimizers. The oracle diagnostic is intentionally not a valid competing initializer. The experiments are normalized CR3BP studies without ephemeris dynamics, mass depletion, eclipse constraints, navigation uncertainty, thrust pointing limits beyond acceleration norm, or operational constraints.

9. Reproducibility

The repository contains Python source, configurations, source-state metadata, generated tables, figures, and result metadata. The catalog-derived pilot command is:

```
py -3.11 scripts\run_experiment.py --config
configs\pilot_replicated.yaml
```

The stronger catalog-derived candidate command is:

```
py -3.11 -m scripts.run_experiment --config
configs/q1_candidate.yaml
```

The feasibility sweep command is:

```
py -3.11 -m scripts.run_feasibility_sweep
```

The catalog multiple-shooting diagnostic artifacts are regenerated without new solves by:

```
py -3.11 scripts\run_catalog_collocation_feasibility.py
--resume --max-cases 0
```

The bounded-projected multiple-shooting phase-shift diagnostic command that reproduces the positive row is:

```
py -3.11 scripts\run_bounded_collocation_benchmark.py
--resume --segments 12 --selected-outages 1
--min-recovery-segments 1 --max-nfev 40
```

The bounded-projected phase-suite command is:

```
py -3.11 scripts\run_bounded_phase_suite.py --resume
```

The compact trapezoidal direct-collocation baseline command is:

```
py -3.11 scripts\run_direct_collocation_baseline.py
```

The targeted catalog feasibility command used in the additional negative test is:

```
py -3.11 -m scripts.run_feasibility_sweep --config
configs/catalog_targeted_feasibility.yaml --transfer-times 4.0
--amax 0.3 --segments 14 --max-nfev 250 --multistart
--random-starts 3 --include-bang-bang --min-recovery-segments 4
```

The non-teacher catalog halo phase-shift benchmark command is:

```
py -3.11 scripts\run_experiment.py --config
configs\q1_phase_shift.yaml
```

The deterministic dense-schedule repair ablation for the same benchmark is:

```
py -3.11 scripts\run_experiment.py --config
configs\q1_phase_shift_repair.yaml
```

The 12-segment duty-cycle-prior phase-shift benchmark command is:

```
py -3.11 scripts\run_experiment.py --config
configs\q1_phase_shift_cardinality.yaml
```

The duty-cycle ablation and all-windows fuel-regularization sweep command is:

```
py -3.11 scripts\run_cardinality_ablation.py
```

The optimized-angle QAOA depth ablation command is:

```
py -3.11 scripts\run_qaoa_depth_ablation.py
--angle-restarts 1 --maxiter 10
```

The teacher-controlled benchmark command is:

```
py -3.11 -m scripts.run_experiment --config
configs/q1_teacher_feasible.yaml
```

The run metadata record Python version, package versions, operating system, configuration, target-state generation, true-evaluation accounting, norm-bounded refinement, and claims limitations. The bounded phase-suite and direct-collocation metadata also record per-row settings fingerprints, configuration hashes, and source-state identifiers so that resumed rows are rejected and recomputed when effective settings change. A scoped artifact manifest with SHA-256 hashes is stored in `data/results/artifact_manifest.json`, and the pinned direct-dependency set is stored in `requirements-lock.txt`. The current run used installed Python 3.11 because creation of a project-local virtual environment stalled on the NAS-backed workspace; this remains a reproducibility weakness until a clean external clone is verified. The test suite checks objective/QUBO finiteness, target-mode metadata, teacher control norm bounds, norm-bounded refinement controls, branch-recovery accounting, direct-collocation control bounds and resume fingerprints, QAOA angle optimization, and table-generation behavior.

10. Conclusion

This paper formulates quantum-assisted robust maneuver initialization for low-thrust cislunar transfers as a binary scheduling problem with missed-thrust penalties embedded in the energy model. The proposed workflow keeps the split between quantum-ready discrete search and classical continuous trajectory optimization explicit. The generated evidence now includes non-teacher catalog halo phase-shift benchmarks that bounded-projected multiple shooting and compact trapezoidal direct collocation solve with selected-branch recovery models in limited cases, showing that the continuous refinement machinery is not limited to teacher-generated targets when acceleration bounds are enforced consistently inside the residual. However, the same benchmark family shows that the tested unrepaired binary schedule samplers can fail to identify a refinable thrust-window structure. A deterministic dense-schedule repair makes the phase-shift case refinable for all samplers, but only by converting their sparse candidates to the same `11111111` all-windows envelope, so it is a diagnostic of encoding sparsity rather than a performance win. A duty-cycle-prior variant gives a more constructive but still limited result: a QUBO-ready active-fraction prior lets surrogate-QUBO simulated annealing and strong classical heuristics find one-coast schedules that remain robustly refinable. Optimized-angle statevector QAOA improves the quantum-ready sampler, with $p = 2$ reaching $9/10$ success in a focused ablation, but it still does not beat surrogate-QUBO simulated annealing. Exhaustive cardinality and fuel-regularization ablations show that the one-coast schedules are one part of a broad high-duty feasible region and are not Pareto-superior to all-windows continuous refinement with stronger fuel regularization. The teacher-controlled benchmark shows that selected-branch robust recovery can be achieved by several initializers, but random search currently has the highest non-oracle success rate and the oracle diagnostic shows that continuous-control initialization can dominate the outcome. The hard NRHO-like to DRO-phase catalog-derived benchmark remains unresolved. The defensible conclusion is therefore not that quantum-assisted initialization is superior today, but that the proposed framework identifies exactly where such a claim would need to be earned: duty-cycle-aware binary encodings that preserve dense low-thrust arcs while adding trajectory-shaping value, stronger continuous-control initialization, stronger surrogate and QAOA generalization, and statistically significant improvement over robust classical initializers on non-constructed benchmarks under equal total evaluation budgets.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

All code and generated artifacts are available in the working repository associated with this draft. The benchmark source states are derived from the NASA/JPL Three-Body Periodic Orbits API and recorded in `data/source_states.json`.

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